

Laboratory Exercise

Network Diagnostics and Infrastructure Documentation Using Command Prompt

Objectives

At the end of this activity, students should be able to:

- Identify network connectivity and communication status.
- Examine routing and DNS information using Command Prompt.
- Interpret basic network diagnostic results.
- Document network infrastructure observations.

Instructions

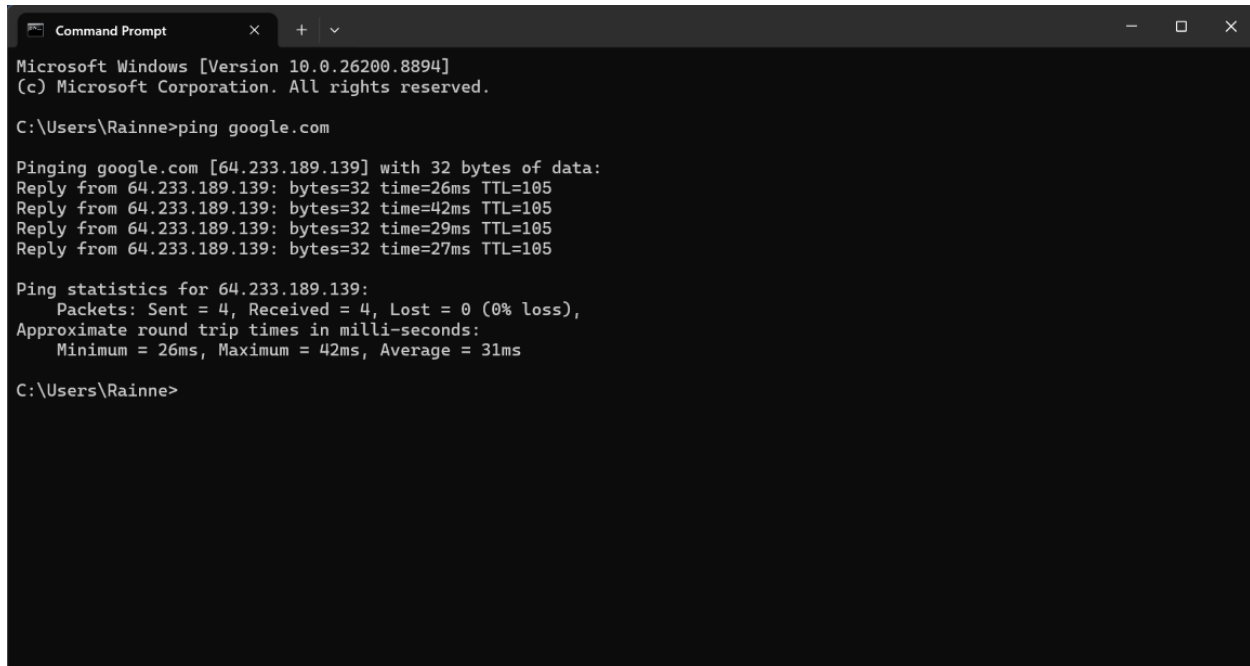
Open Command Prompt (CMD) and execute the following commands. Record your observations.

Command	Purpose
ping	Tests network connectivity
tracert	Displays the path to a destination
nslookup	Resolves domain names to IP addresses
arp -a	Displays ARP cache (IP and MAC addresses)
netstat -an	Displays active network connections

Test Internet Connectivity

Command

ping google.com

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The text inside shows the Windows version (10.0.26200.8894) and the user's location (C:\Users\Rainne). The command 'ping google.com' has been entered and executed. The output shows four successful replies from the IP address 64.233.189.139 with varying response times (26ms, 42ms, 29ms, 27ms) and a TTL of 105. Ping statistics are also displayed, showing 4 packets sent, 4 received, 0% loss, and an average response time of 31ms.

```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rainne>ping google.com

Pinging google.com [64.233.189.139] with 32 bytes of data:
Reply from 64.233.189.139: bytes=32 time=26ms TTL=105
Reply from 64.233.189.139: bytes=32 time=42ms TTL=105
Reply from 64.233.189.139: bytes=32 time=29ms TTL=105
Reply from 64.233.189.139: bytes=32 time=27ms TTL=105

Ping statistics for 64.233.189.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 42ms, Average = 31ms

C:\Users\Rainne>
```

Record the following:

Reply Received:

Reply from 64.233.189.139: bytes=32 time=26ms TTL=105

Reply from 64.233.189.139: bytes=32 time=42ms TTL=105

Reply from 64.233.189.139: bytes=32 time=29ms TTL=105

Reply from 64.233.189.139: bytes=32 time=27ms TTL=105

Packets Sent: Sent = 4

Packets Lost: Lost = 0

Average Response Time: Average = 31ms

Trace the Network Path

Command

tracert google.com

```
Command Prompt
Tracing route to google.com [64.233.189.139]
over a maximum of 30 hops:

  1  14 ms  1 ms  1 ms  dev.opt [192.168.100.1]
  2   6 ms  5 ms  3 ms  10.97.184.1
  3   3 ms  3 ms  3 ms  161.49.2.121.convergeict.com [161.49.2.121]
  4   4 ms  3 ms  3 ms  161.49.7.234.convergeict.com [161.49.7.234]
  5   4 ms  3 ms  3 ms  161.49.4.41.convergeict.com [161.49.4.41]
  6  15 ms  17 ms  3 ms  161.49.6.144.convergeict.com [161.49.6.144]
  7  15 ms  18 ms  3 ms  142.250.172.244
  8   6 ms  3 ms  3 ms  142.251.251.39
  9   4 ms  7 ms  4 ms  142.251.192.82
 10   6 ms  5 ms  5 ms  142.251.49.169
 11  *      *      *      Request timed out.
 12 28 ms  26 ms  29 ms  209.85.247.44
 13 29 ms  29 ms  30 ms  74.125.253.85
 14  *      *      *      Request timed out.
 15  *      *      *      Request timed out.
 16  *      *      *      Request timed out.
 17  *      *      *      Request timed out.
 18  *      *      *      Request timed out.
 19  *      *      *      Request timed out.
 20  *      *      *      Request timed out.
 21  *      *      *      Request timed out.
 22  *      *      *      Request timed out.
 23 28 ms  25 ms  25 ms  tl-in-f139.1e100.net [64.233.189.139]

Trace complete.

C:\Users\Rainne>
```

Record:

Number of hops reached: 23

Did the trace successfully reach Google?

☒ Yes

☐ No

DNS Lookup

Command

nslookup google.com

```
Command Prompt
8      6 ms    3 ms    3 ms    142.251.251.39
9      4 ms    7 ms    4 ms    142.251.192.82
10     6 ms    5 ms    5 ms    142.251.49.169
11     *      *      *      Request timed out.
12    28 ms    26 ms    29 ms    209.85.247.44
13    29 ms    29 ms    30 ms    74.125.253.85
14     *      *      *      Request timed out.
15     *      *      *      Request timed out.
16     *      *      *      Request timed out.
17     *      *      *      Request timed out.
18     *      *      *      Request timed out.
19     *      *      *      Request timed out.
20     *      *      *      Request timed out.
21     *      *      *      Request timed out.
22     *      *      *      Request timed out.
23    28 ms    25 ms    25 ms    tl-in-f139.1e100.net [64.233.189.139]

Trace complete.

C:\Users\Rainne>nslookup google.com
Server: dev.opt
Address: 192.168.100.1

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4017:806::200e
           142.250.205.238

C:\Users\Rainne>
```

Record:

DNS Server: dev.opt

Google IP Address: 2404:6800:4017:806::200e

142.250.205.238

View Network Devices

Command

arp -a

```
Command Prompt
21 * * * Request timed out.
22 * * * Request timed out.
23 28 ms 25 ms 25 ms tl-in-f139.1e100.net [64.233.189.139]

Trace complete.

C:\Users\Rainne>nslookup google.com
Server: dev.opt
Address: 192.168.100.1

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4017:806::200e
          142.250.205.238

C:\Users\Rainne>arp -a

Interface: 192.168.100.7 --- 0xa
Internet Address Physical Address Type
192.168.100.1 7c-39-85-c6-07-ac dynamic
192.168.100.3 7a-8e-93-ab-c5-2b dynamic
192.168.100.255 ff-ff-ff-ff-ff-ff static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.251 01-00-5e-00-00-fb static
224.0.0.252 01-00-5e-00-00-fc static
239.255.255.250 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static

C:\Users\Rainne>
```

Complete the table.

Internet Address / IP Address Physical / MAC Address Type

192.168.100.1	7c-39-85-c6-07-ac	Dynamic
192.168.100.3	7a-8e-93-ab-c5-2b	Dynamic
192.168.100.255	ff-ff-ff-ff-ff-ff	Static
224.0.0.22	01-00-5e-00-00-16	Static
224.0.0.251	01-00-5e-00-00-fb	Static
224.0.0.252	01-00-5e-00-00-fc	Static
239.255.255.250	01-00-5e-7f-ff-fa	Static
255.255.255.255	ff-ff-ff-ff-ff-ff	Static

View Active Network Connections

Command

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING
TCP	0.0.0.0:28252	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING
TCP	127.0.0.1:24830	0.0.0.0:0	LISTENING
TCP	127.0.0.1:55070	127.0.0.1:55071	ESTABLISHED
TCP	127.0.0.1:55071	127.0.0.1:55070	ESTABLISHED
TCP	192.168.100.7:139	0.0.0.0:0	LISTENING
TCP	192.168.100.7:49169	142.250.205.234:443	ESTABLISHED
TCP	192.168.100.7:49491	172.172.255.218:443	ESTABLISHED
TCP	192.168.100.7:49671	52.159.82.83:443	CLOSE_WAIT
TCP	192.168.100.7:49968	172.217.27.142:443	ESTABLISHED
TCP	192.168.100.7:50013	54.157.224.50:443	ESTABLISHED
TCP	192.168.100.7:50953	35.190.80.1:443	ESTABLISHED
TCP	192.168.100.7:50980	74.125.204.94:443	ESTABLISHED
TCP	192.168.100.7:51148	142.250.205.238:443	ESTABLISHED
TCP	192.168.100.7:51173	142.251.220.129:443	TIME_WAIT
TCP	192.168.100.7:51694	161.49.29.84:443	TIME_WAIT
TCP	192.168.100.7:51857	74.125.204.94:443	TIME_WAIT
TCP	192.168.100.7:51918	52.110.8.131:443	ESTABLISHED
TCP	192.168.100.7:52124	108.177.125.188:5228	ESTABLISHED

TCP	192.168.100.7:52347	172.64.155.209:443	ESTABLISHED
TCP	192.168.100.7:52460	172.217.113.4:443	ESTABLISHED
TCP	192.168.100.7:52526	172.64.144.52:443	ESTABLISHED
TCP	192.168.100.7:52527	161.49.36.81:443	ESTABLISHED
TCP	192.168.100.7:52789	100.57.239.57:443	ESTABLISHED
TCP	192.168.100.7:52849	172.217.27.142:443	ESTABLISHED
TCP	192.168.100.7:52860	52.111.240.76:443	ESTABLISHED
TCP	192.168.100.7:52927	57.144.228.141:443	ESTABLISHED
TCP	192.168.100.7:53022	108.177.97.94:443	ESTABLISHED
TCP	192.168.100.7:53030	172.64.146.98:443	ESTABLISHED
TCP	192.168.100.7:53158	34.195.103.245:443	ESTABLISHED
TCP	192.168.100.7:53275	3.213.144.91:443	ESTABLISHED
TCP	192.168.100.7:53357	104.18.37.228:443	ESTABLISHED
TCP	192.168.100.7:53603	52.111.240.76:443	ESTABLISHED
TCP	192.168.100.7:53761	57.144.184.145:443	ESTABLISHED
TCP	192.168.100.7:55067	128.116.97.3:443	ESTABLISHED
TCP	192.168.100.7:55076	128.116.97.3:443	ESTABLISHED
TCP	192.168.100.7:55078	128.116.97.3:443	ESTABLISHED
TCP	192.168.100.7:55656	135.233.95.135:443	TIME_WAIT
TCP	192.168.100.7:55705	142.250.207.42:443	TIME_WAIT
TCP	192.168.100.7:55707	57.155.104.224:443	TIME_WAIT
TCP	192.168.100.7:55708	72.154.7.248:443	TIME_WAIT
TCP	192.168.100.7:55709	72.153.5.132:443	ESTABLISHED
TCP	192.168.100.7:55710	27.49.26.55:443	ESTABLISHED
TCP	192.168.100.7:55712	27.49.26.55:443	ESTABLISHED
TCP	192.168.100.7:56056	75.101.143.69:443	ESTABLISHED
TCP	192.168.100.7:56386	172.217.27.65:443	ESTABLISHED
TCP	192.168.100.7:56602	57.144.228.141:443	ESTABLISHED
TCP	192.168.100.7:56828	57.144.184.145:443	ESTABLISHED
TCP	192.168.100.7:57475	104.18.42.153:443	TIME_WAIT

TCP	192.168.100.7:58024	74.125.204.94:443	ESTABLISHED
TCP	192.168.100.7:58373	161.49.29.209:443	TIME_WAIT
TCP	192.168.100.7:58534	52.111.240.76:443	ESTABLISHED
TCP	192.168.100.7:58547	142.251.220.138:443	ESTABLISHED
TCP	192.168.100.7:59183	52.123.224.2:443	TIME_WAIT
TCP	192.168.100.7:59743	57.144.184.145:443	ESTABLISHED
TCP	192.168.100.7:59769	3.213.144.91:443	ESTABLISHED
TCP	192.168.100.7:59895	172.217.117.4:443	ESTABLISHED
TCP	192.168.100.7:60116	52.110.8.151:443	ESTABLISHED
TCP	192.168.100.7:60993	52.110.8.153:443	ESTABLISHED
TCP	192.168.100.7:60994	52.110.8.153:443	ESTABLISHED
TCP	192.168.100.7:60995	52.110.8.128:443	ESTABLISHED
TCP	192.168.100.7:61425	3.170.249.95:443	ESTABLISHED
TCP	192.168.100.7:61643	104.18.39.21:443	ESTABLISHED
TCP	192.168.100.7:61863	142.250.207.46:443	TIME_WAIT
TCP	192.168.100.7:62905	142.251.220.129:443	TIME_WAIT
TCP	192.168.100.7:63083	57.144.228.141:443	ESTABLISHED
TCP	192.168.100.7:63240	57.144.183.134:443	ESTABLISHED
TCP	192.168.100.7:64321	3.170.252.34:443	ESTABLISHED
TCP	192.168.100.7:64345	13.107.226.39:443	ESTABLISHED
TCP	192.168.100.7:64370	142.250.157.95:443	ESTABLISHED
TCP	192.168.100.7:64984	57.144.184.145:443	ESTABLISHED
TCP	192.168.100.7:65086	100.57.239.57:443	ESTABLISHED
TCP	[::]:135	:::0	LISTENING
TCP	[::]:445	:::0	LISTENING
TCP	[::]:49664	:::0	LISTENING
TCP	[::]:49665	:::0	LISTENING
TCP	[::]:49666	:::0	LISTENING
TCP	[::]:49667	:::0	LISTENING
TCP	[::]:49668	:::0	LISTENING

TCP	[::]:49670	:::0	LISTENING
TCP	[::1]:49669	:::0	LISTENING
UDP	0.0.0.0:5050	::*	
UDP	0.0.0.0:5353	::*	
UDP	0.0.0.0:5353	::*	
UDP	0.0.0.0:5353	::*	
UDP	0.0.0.0:5353	::*	
UDP	0.0.0.0:5353	::*	
UDP	0.0.0.0:5355	::*	
UDP	0.0.0.0:28252	::*	
UDP	0.0.0.0:50259	64.233.187.113:443	
UDP	0.0.0.0:50686	142.251.157.119:443	
UDP	0.0.0.0:50875	::*	
UDP	0.0.0.0:51489	142.250.205.234:443	
UDP	0.0.0.0:52778	74.125.204.94:443	
UDP	0.0.0.0:54357	142.251.220.138:443	
UDP	0.0.0.0:54501	108.177.97.94:443	
UDP	0.0.0.0:55499	172.217.27.142:443	
UDP	0.0.0.0:55794	142.250.205.238:443	
UDP	0.0.0.0:57765	172.217.26.78:443	
UDP	0.0.0.0:59169	74.125.204.94:443	
UDP	0.0.0.0:61411	64.233.187.113:443	
UDP	0.0.0.0:62007	57.144.182.141:443	
UDP	0.0.0.0:62162	172.217.27.142:443	
UDP	0.0.0.0:62310	172.217.27.138:443	
UDP	0.0.0.0:63043	108.177.97.94:443	
UDP	0.0.0.0:64392	74.125.204.94:443	
UDP	0.0.0.0:65046	142.251.220.138:443	
UDP	127.0.0.1:1900	::*	
UDP	127.0.0.1:49664	127.0.0.1:49664	

UDP	127.0.0.1:49664	127.0.0.1:49664	
UDP	127.0.0.1:55819	::*	
UDP	192.168.100.7:137	::*	
UDP	192.168.100.7:138	::*	
UDP	192.168.100.7:1900	::*	
UDP	192.168.100.7:2177	::*	
UDP	192.168.100.7:55818	::*	
UDP	[::]:5353	::*	
UDP	[::]:5353	::*	
UDP	[::]:5353	::*	
UDP	[::]:5355	::*	
UDP	[::]:50875	::*	
UDP	[::1]:1900	::*	
UDP	[::1]:55817	::*	
UDP	[fe80::b6d6:5f6f:8165:d7aa%10]:1900	::*	
UDP	[fe80::b6d6:5f6f:8165:d7aa%10]:2177	::*	
UDP	[fe80::b6d6:5f6f:8165:d7aa%10]:55816	::*	

netstat -an

Record:

Number of Active Connections: 42

Any Listening Ports? There are many listening ports